Topic 1: Problem solving

Students are expected to develop a set of computational thinking skills that enable them to understand how computer systems work and design, implement and analyse algorithms for solving problems.

Students should be given repeated opportunities to tackle computational problems of various sorts, including some substantial problem-solving tasks.

Subject content	Students should:
1.1 Decomposition	1.1.1 Be able to analyse a problem, investigate requirements [inputs, outputs, processing], specify success criteria and design solutions
	1.1.2 Be able to decompose a problem into smaller sub-problems
1.2. Algorithms	1.2.1 Understand what an algorithm is and what algorithms are used for
	1.2.2 Be able to create an algorithm to solve a particular problem making use of programming constructs [sequence, selection, repetition]
	1.2.3 Be able to describe the purpose of a given algorithm and explain how a simple algorithm works
	1.2.4 Be able to identify the correct output of an algorithm for a given set of data
	1.2.5 Be able to identify and correct errors in algorithms
	1.2.6 Be able to code an algorithm into a high-level language (from pseudocode, flow charts, structured English or written descriptions)*
	1.2.7 Understand how standard algorithms [quick sort, bubble sort, selection sort, linear search, binary search, breadth first search, depth first search, maximum/minimum, mean, count] work
	1.2.8 Understand factors that affect the efficiency of an algorithm [worst case, average case, best case]

* See *Appendix 2* for a key definitions related to this content.